Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

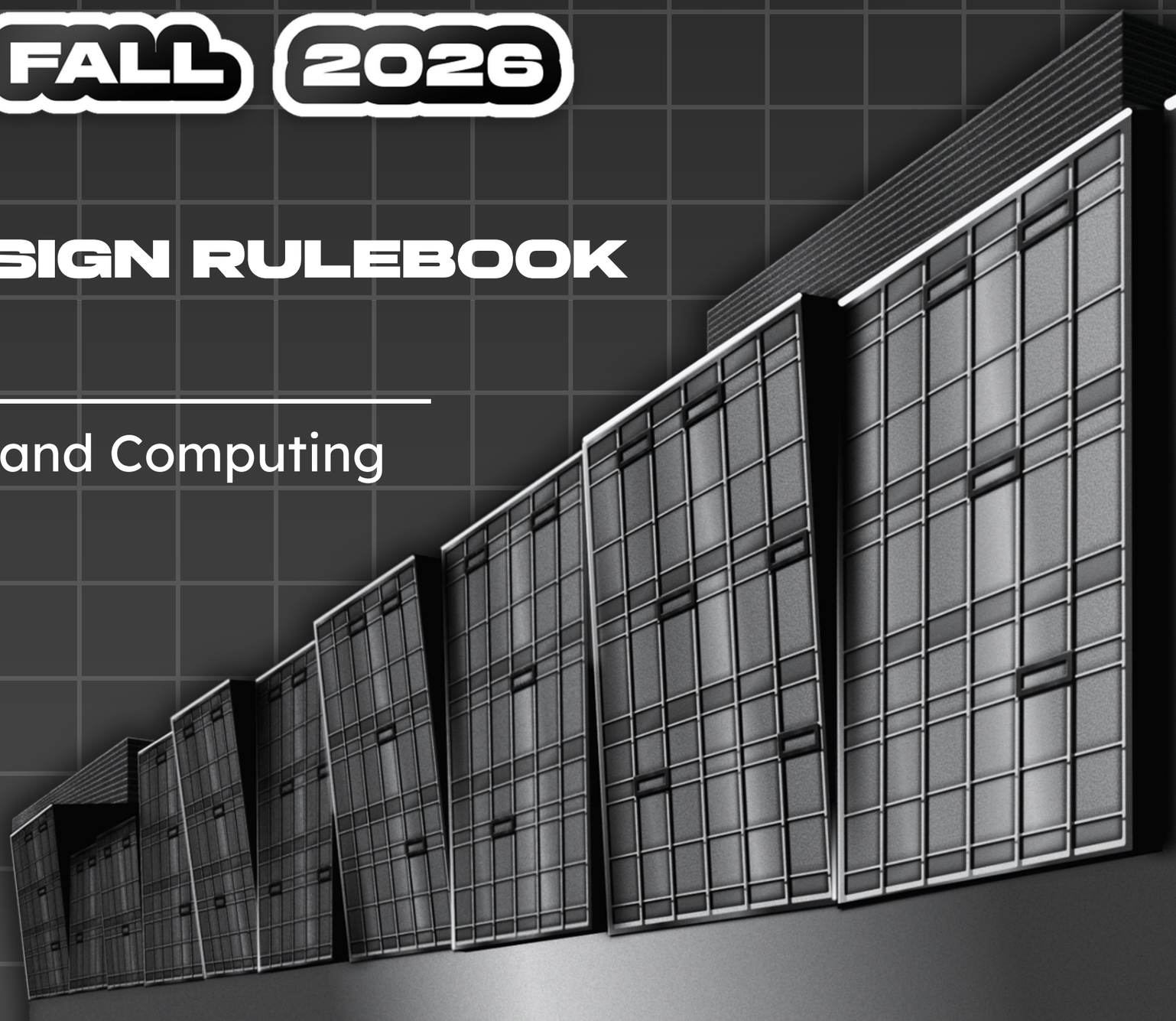
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

INNOVATIVE DESIGN RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



1.1 Goal

The MEC 2026 Innovative Design Competition challenges teams to develop an innovative engineering proposal for an Urban Food Security District in Scarborough, Toronto. Competitors must follow the official competition package, including its required facilities, design pillars, land-use constraints, and budget requirements. Teams will prepare their designs before competition day and present their proposed solutions to a panel of judges. The designs must be new and innovative and address a void in society.

1.2 Team Composition

Each Innovative Design team must comprise five (5) to six (6) registered competitors. Should the presenting team be a subset of the project team, the remaining members of the project team must sign a letter authorizing the presentation of the project for it to be eligible for the competition.

1.3 Project Requirements

- All teams must develop an original engineering design proposal based on the official competition problem statement and requirements provided by the Competition Organizers.
- Teams may choose their own engineering approaches, technologies, and design solutions, provided they follow the specified constraints and submission requirements. All proposed work must be original and supported by appropriate research and engineering justification.

1.4 Language

The design summary, presentation display, and presentation can be done in either English or French but must be consistent in language.

1.5 Resources

1.5.1 Personnel Required

1.5.1.1 Judges

A minimum of three (3) judges (and in any excess, an odd number of judges) are required to assess the problem-solving abilities, proposed solution, communication skills and team dynamics of the competitors. Judges in this category should come from a variety of backgrounds including communications, sales, and technical or consulting engineering experience related to the topic.

1.5.1.2 Competition Lead

The Competition Lead is responsible for the logistics and implementation of the Innovative Design competition. The Competition Lead must be available to competitors and judges at any time for questions and requests.

1.5.1.3 Official Timekeeper

The Official Timekeeper is typically the Competition Lead who enforces the time limits and the rules governing the displays.

1.5.2 Equipment Provided by MEC

The following equipment will be made available to all competitors during the competition time period:

- A table to display presentation
- Electrical power supply for each team
- Internet source

1.5.3 Allowed External Resources

There is no restriction on the resources which competitors are allowed to bring to the competition.

1.6 Procedure/Timeline

1.6.1 Pre-Competition

Prior to MEC	Design Summary and Presentation Submission: Teams must submit a one (1) page Design Summary and their presentation slides in PDF format by the official submission deadline communicated by the Competition Organizers. Further requirements are outlined in Section 1.6.5 (Deliverables).
Prior to Presentations	Competition Briefing: The Competition Coordinators will provide competitors with information regarding the competition schedule, presentation procedures, judging format, and applicable rules. Competitors will have an opportunity to ask questions about competition procedures.

1.6.2 Judging

The judges, either individually or in groups of not more than three (3), will walk among the displays during the judging period. The competitors must give a fifteen (15) minute presentation, followed by a five (5) minute question period, and a five (5) minute Supply Chain Curvball question (This question will remain undisclosed until the day of the competition), at their displays. They must use only the resources at their display to illustrate their work. Competitors will be required to give their presentation more than once as each judge must see all displays, but not at the same time.

1.6.3 Timekeeping

The following rules will be adhered with respect to timekeeping:

- Time is not halted when a judge asks a question during the presentation
- The remaining time must be indicated to the competitors ten (10) minutes, five (5) minutes, and one (1) minute before the end of the allotted time for the presentation
- A visual countdown must be given during the last thirty (30) seconds of the presentation
- Presenters will be asked to stop their prepared presentation when the fifteen (15) minute presentation limit has been reached. The five (5) minute question-and-answer period and the five (5) minute Supply Chain Curveball period will be timed separately.
- Time to both ask and answer questions during the question period is counted

1.6.4 Presentation Order

- Presentation order shall be determined randomly
- Presentation order shall be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement
- Teams are not allowed to switch places in the presentation order
- The judges may view the presentations either individually or in small groups of up to three (3) people
- Competitors will be required to give their presentation more than once
- The order should allow all the judges to visit all the displays
- The order shall prevent the judges from visiting the displays in the same order

1.6.5 Deliverables

1.6.5.1 Design Summary

Each team must submit a maximum one-page Design Summary in PDF format by Friday, November 6, 2026, at 11:59 PM Eastern Time, through the official submission form. The purpose of this summary is for the judges and organizers to be prepared ahead of time with regard to the topics. This summary will follow conventional formatting characteristics (12-point front, 1.5 line spacing, 1” paper margins) and is limited to a maximum of one (1) page. The summary must include, but is not limited to:

- Team Name / Alias
- Team Members
- Design topics and descriptions

1.6.5.2 Presentation

Each team must submit its presentation slides in PDF format through the official submission form by Friday, November 6, 2026, at 11:59 PM Eastern Time. Teams are not required to provide printed copies unless specifically instructed by the Competition Coordinators.

1.7 Assessment and Judging

- The panel must have an odd number of judges
- The panel must comprise a minimum of three (3) judges. There is no maximum number of judges
- Feedback forms shall be provided to each team following the announcement of winners given prior to the end of MEC

1.7.1 Judging Matrix

Innovative Design Judging Matrix

Project	Space Efficiency & Functional Design /30 Environmental Resilience /15 Economic Sustainability /10 Innovation & Engineering Creativity /15	**/70**
Design Summary	Content of the Overview is Clear and Concise /5 Engineering Justification for all Major Design Decisions /5	**/10**
Presentation	Clear Communication when explaining the Proposed Design /5 Strong Engineering Knowledge and provides Justified Answers to Judges' Questions /5 Presentation Visuals & Time Management /5 Provides a Clear and Justified answer to a Supply Chain disruption Question /5	**/20**
Total		**/100**

1.7.2 Penalty Matrix

Penalties

Plagiarism	-50
Insufficient Citation	-50
Absent Team Member(s)	-25
Late submission of the Design Summary or Presentation Slides: Five (5) points deducted for each day late, up to a maximum deduction of fifty (50) points.	Up to -50
For Every Minute the Presentation Exceeds the Time Limit Two (2) Points Deduction	Up to -50
For Every Three (3) Grammar or Spelling Mistake in the Abstract One (1) Point Deduction	Up to -50

Any use and or suspected use of AI will be investigated and may result in an immediate disqualification from the competition.